

CERTIFIED ZEROS OF THE TRANSFER-OPERATOR DETERMINANT FOR THE HECKE TRIANGLE GROUP G_5

[AUTHORS]

ABSTRACT. The plus-sector transfer-operator determinant $\det(1 - L_{s,+})$ of the Hecke triangle group G_5 has two zeros s_1 and s_2 , each of multiplicity one, confined by interval arithmetic to explicit boxes of coordinate half-width 10^{-6} whose closed real-part intervals are disjoint. The computation upgrades the numerical zeros recorded by Strömberg for $q = 3, 4, 5$, which that work explicitly described as non-rigorous, to a certified argument-principle standing: each box is covered by a closed contour on which a finite-section Fredholm determinant is enclosed in Arb ball arithmetic with directed rounding, the enclosure excludes zero, and the certified winding number is 1.

A published factorization of the Selberg zeta function of G_5 , together with a published divisor theorem for cofinite orbifolds, would promote each certified zero of $\det(1 - L_{s,+})$ to a zero of Z_{G_5} of multiplicity at least one and, after the scattering functional equation, to a zero of the scalar scattering determinant φ_5 in the strip $1/2 < \Re s < 1$. That promotion is stated as a conditional proposition, with every joint labelled certified, cited, or open. No infinitude law for scattering zeros is claimed.

1. INTRODUCTION

For the modular group $\mathrm{PSL}(2, \mathbb{Z})$ the scattering determinant of the cusp is an explicit ratio of completed Riemann zeta functions, so its nonreal zeros and poles are governed by the zeros of ζ and, on the Riemann hypothesis, lie on the critical line $\Re s = 1/2$. The same mechanism operates for every arithmetic Fuchsian group: number-theoretic L -functions supply the scattering data and pin its zeros to the line. For a non-arithmetic group no such L -function mechanism exists, and the Phillips–Sarnak theory of cusp-form dissolution predicts that spectral data should instead migrate into genuinely off-line resonances. Deciding whether the resonances of a *specific* non-arithmetic group actually leave the line is then a quantitative question about one meromorphic function, and the smallest natural test case is the Hecke triangle group G_5 , the first non-arithmetic member of the simplest one-parameter family of Fuchsian groups. Strömberg computed numerical candidates for its lowest resonances and stated explicitly that the computation was not rigorous. This paper closes that gap for two of them: each is confined to an explicit box by interval arithmetic with a certified winding number, and the two boxes have disjoint real parts. The certification is unconditional; what the two certified zeros do and do not imply about scattering resonances is stated as a separate, fully labelled conditional proposition.

Let $q \geq 3$ be an integer. The Hecke triangle group G_q is the subgroup of $\mathrm{PSL}(2, \mathbb{R})$ generated by the involution $z \mapsto -1/z$ and the translation $z \mapsto z + \lambda_q$, where $\lambda_q = 2 \cos(\pi/q)$. The quotient $X_q = G_q \backslash \mathbb{H}$ is a cofinite hyperbolic orbifold with one cusp and two elliptic points, of orders 2 and q . For $q = 5$ one has $\lambda_5 = 2 \cos(\pi/5)$, and G_5 is non-arithmetic [19]. Write Z_{G_q} for the Selberg zeta function of G_q , and write $L_{s,\pm}$ for the plus/minus sectors of the Mayer–Mühlenbruch–Strömberg transfer operator of G_q [11]. The object certified in this paper is the plus-sector Fredholm determinant $\det(1 - L_{s,+})$.

Date: DRAFT — not for distribution.

2020 Mathematics Subject Classification. Primary 11F72, 37C30; Secondary 11Y35, 58J50, 30C15.

Key words and phrases. Hecke triangle group, transfer operator, Selberg zeta function, scattering determinant, interval arithmetic, certified computation.

Definition 1.1. A *resonance* of X_q is a pole of the meromorphically continued scalar scattering determinant φ_q of the trivial character. Under the divisor theorem recalled in Theorem 2.1 below, every nonreal zero of Z_{G_q} in the strip $0 < \Re s < 1/2$ is such a pole, with matching multiplicity, and conversely.

Strömberg [18] computed numerical zeros of Z_{G_q} for $q = 3, 4, 5$ from truncated transfer-operator determinants and stated that the computation is not rigorous. The present paper supplies interval certificates for two zeros of $\det(1 - L_{s,+})$ at $q = 5$. Winding number 1 on a contour that excludes zero certifies a simple zero of $\det(1 - L_{s,+})$ in each box. It does not, by itself, certify simplicity of a zero of Z_{G_5} : in the factorization of [11, Theorem 6.4],

$$(1) \quad Z_{G_q}(s) = \frac{\det(1 - L_{s,+}) \det(1 - L_{s,-})}{\det(1 - K_s)},$$

the minus-sector factor may vanish at the same point. Accordingly Z_{G_5} is asserted only to vanish with multiplicity at least that of the plus factor, and only after the joints of Proposition 5.1 that lie outside the certificates are granted.

The paper does not prove that every finite q has infinitely many off-critical-line scattering zeros. That fixed- q growth statement is discussed only in §6, as an open comparison with Selberg, Hejhal, and Kelmer.

The contents are as follows. Section 2 records the cited analytic apparatus. Section 3 describes the interval certificates. Section 4 states the unconditional two-pin theorem. Section 5 enumerates the thirteen joints from the plus-factor zeros to a scattering consequence, labelled certified, cited, or open. Section 6 compares prior theorems with the present certificates. Sections 7 and 8 record the Lean artifacts at their actual strength and the data archive.

Notation.

Symbol	Meaning
$q \geq 3$	integer elliptic order
G_q	Hecke triangle group $\langle z \mapsto -1/z, z \mapsto z + \lambda_q \rangle$
λ_q	$2 \cos(\pi/q)$
X_q	cofinite orbifold $G_q \backslash \mathbb{H}$
Z_{G_q}	Selberg zeta function of G_q
$L_{s,\pm}$	plus/minus Mayer–Mühlenbruch–Strömberg transfer operators
$\det(1 - L_{s,+})$	plus-sector Fredholm determinant
K_s	MMS correction operator in the factorization (1)
φ_q	scalar trivial-character scattering determinant of X_q
Ω^*	$\{\Re s > 1/2\} \cup \{\Re s > 0, \Im s > 1\}$
$\text{Box}_1, \text{Box}_2$	certified squares of half-width 10^{-6} , §3

2. BACKGROUND

2.1. Hecke triangle groups. For integer $q \geq 3$, the group G_q is a cofinite discrete subgroup of $\text{PSL}(2, \mathbb{R})$ with signature $(0; 2, q, \infty)$. It is arithmetic if and only if $q \in \{3, 4, 6\}$ [19]. The surface X_q has one cusp, so the scattering matrix of the trivial character is 1×1 ; its determinant is a scalar meromorphic function φ_q . We work throughout with the conjugated normalization in which the cusp has width 1, as in Hejhal [6, Ch. 7]. Changing the width multiplies φ_q by a zero-free factor of the form c^{1-2s} , which preserves zeros, poles, and the functional equation $\varphi_q(s)\varphi_q(1-s) = 1$.

2.2. Transfer operators. Mayer, Mühlenbruch, and Strömberg construct, for each integer $q \geq 3$, a nuclear transfer operator on a finite direct sum of Hardy spaces of disks, together with a $\pm$ decomposition under a CF-reflection involution [11]. For odd q the reduced plus-sector operator $L_{s,+}$ is the operator displayed by the odd- q incidence formulas of that paper (the three-row reduced block structure). At $q = 5$ one has $h_q = 1$ and $\kappa_q = 3$. The heading printed above [11, (34)] reads

“ $q = 2h_q + 3 > 5$ ”, while the general odd- q incidence formulas, Lemma 6.3, Theorem 6.4, and the explicit $q = 5$ functional equation of the same paper are stated for odd $q \geq 5$. We use the general incidence formula at $h_q = 1$, and we record the heading conflict as an open joint in Proposition 5.1 (J8).

The Fredholm determinant $\det(1 - L_{s,+})$ is entire in s on the domain

$$(2) \quad \Omega^* = \{\Re s > 1/2\} \cup \{\Re s > 0, \Im s > 1\}$$

on which the Hilbert-space realization used in the certificates and the Banach-space realization of [11] are identified by the transport discussed as joint (J6) below. Both certified boxes of Theorem 4.1 lie in the second component of Ω^* .

Möller–Pohl [12] and Adam–Pohl [1] give closely related Fredholm-determinant representations of Z_{G_q} for every cofinite Hecke triangle group, via period functions and slow/fast transfer operators. We compute with the MMS plus-sector operator, which is the operator implemented by the archived engine.

2.3. Selberg zeta function. The Selberg zeta function of a cofinite Fuchsian group is the Euler product over primitive closed geodesics

$$(3) \quad Z_\Gamma(s) = \prod_{[\gamma]} \prod_{k=0}^{\infty} (1 - N(\gamma)^{-(s+k)}),$$

absolutely convergent for $\Re s > 1$ and meromorphic on $\mathbb{C}$. For $\Gamma = G_q$ we write Z_{G_q} . The factorization (1) is [11, Theorem 6.4]: it identifies Z_{G_q} with a quotient of three Fredholm determinants, of which $\det(1 - K_s)$ is an explicit entire correction whose zeros form the lattice

$$(4) \quad s = -n + i\pi k/a_q, \quad n \in \mathbb{N} \cup \{0\}, \quad k \in \mathbb{Z},$$

hence all have real part ≤ 0 . Both boxes of Theorem 4.1 satisfy $\Re s \geq 0.41054273549473627 > 0$, so they are disjoint from this lattice.

[11, Theorem 4.10] states that the MMS determinants $\det(1 - L_{s,\pm})$ are meromorphic with poles only at the real points $s = (1 - k)/2$. Both certified boxes have imaginary part strictly larger than 5, so $\det(1 - L_{s,-})$ is holomorphic on a neighbourhood of each box.

2.4. The FJS divisor theorem. Friedman, Jorgenson, and Smajlović [3] determine the divisors of the scattering determinant and of the Selberg zeta function for a cofinite Fuchsian group, including orbifolds, with a finite-dimensional unitary representation. The scalar trivial-character, one-cusp specialization is $h = 1$, $\chi = 1$, and degree of singularity $k = 1$. (Immediately after defining $k = \sum_j \dim V_j$, the source prints a colliding “ $k = 0$ ” in a sentence that evidently uses k for automorphic weight; the divisor formulas retain the degree-of-singularity parameter. The collision does not change the set-level classification used here.)

Theorem 2.1 (Friedman–Jorgenson–Smajlović, §2.4–2.5). *Let Γ be cofinite, possibly with elliptic elements, with one cusp and trivial character, and let φ be its scalar scattering determinant. Then φ continues meromorphically to $\mathbb{C}$ and satisfies $\varphi(s)\varphi(1-s) = 1$. The nonreal zeros of Z_Γ in the strip $0 < \Re s < 1/2$ are exactly the points $1 - \rho$, $1 - \bar{\rho}$ where ρ is a zero of φ with $\Re \rho > 1/2$ and $\Im \rho > 0$, with matching multiplicity. Every identity, elliptic, and parabolic trivial divisor of Z_Γ is real, as is every discrete-spectrum zero.*

The statement is the $h = 1$, $k = 1$ case of [3, §2.5, items 1–7], following Venkov [20] and Hejhal [6, p. 499]. Item 6 of that list is the only source of nonreal off-line zeros of Z_Γ . The gamma and G_1 factors in the completed zeta $Z_+(s) = Z(s)/(G_1(s)\Gamma(s-1/2)^k)$ have real divisors and meet neither box of Theorem 4.1.

Jorgenson–Smajlović [9] supply adjacent cofinite-orbifold context for the Selberg zeta derivative; we do not consume that paper in any certificate.

3. CERTIFICATION METHOD

The certificates are computer-assisted theorems in Arb ball arithmetic [8], accessed through python-flint, at 384 bits. Directed rounding is the Arb midpoint-radius enclosure: every recorded interval is an outward rounding of the true value of the indicated quantity. We describe exactly what the receipts certify.

3.1. Boxes and contours. Each pin is a closed coordinate square of half-width 10^{-6} . Write Box_1 for the square centred at

$$0.4538951800749447 + 5.7635372417301305 i$$

and Box_2 for the square centred at

$$0.41054373549473627 + 7.81976824701551188 i.$$

The boundary ∂Box_i is covered by 192 base straight arcs (48 per edge), oriented counterclockwise. Adaptive bisection, to maximum depth 8, refines any base arc on which the enclosure tests below fail. The first pin uses finite-section rank $N = 160$ and accepts 284 closed subarcs. The second pin uses $N = 288$ and accepts 452 closed subarcs across sixteen chunks of twelve base arcs, merged in `chunk_receipts/S2_MERGED_CONTOUR_RECEIPT.json`.

3.2. Finite-section determinant. Let P_N be the orthogonal projection onto the span of the first N basis vectors in each Hardy component, and write $D_N(s) = \det(I - P_N L_{s,+} P_N)$ for the finite-section determinant. The restriction identity and the Sylvester identity give $D_N(s) = \det_H(I - L_{s,+} P_N)$. On each closed subarc A with midpoint s_0 and radius r , Arb inversion of $I - M(s)$ over A produces a bound $H \geq \sup_A |\text{tr}((I - M(s))^{-1} M'(s))|$. Jacobi's formula then yields $|D'_N(s)| \leq H |D_N(s)|$. Writing rH for the product of r and this H , the gate $rH < 1$ produces an explicit radius G such that the disk $D_N(s_0) + \overline{B}(0, rG)$ contains $D_N(s)$ for every $s \in A$. This Taylor enclosure is recorded per leaf.

3.3. Tail, inflation, and homotopy. A Gohberg–Krein/Simon determinant comparison [17, Theorem 4.2] supplies a uniform bound

$$\left| \det_H(I - L_{s,+}) - D_N(s) \right| \leq F_R := T_{\text{tail}}(N) \exp(1 + 2B_{\text{same}}(N))$$

on the whole closed box, once column-norm bounds for $L_{s,+}$ and $L_{s,+} P_N$ and an input-tail bound $T_{\text{tail}}(N)$ are enclosed. The receipts record these constants. For the second pin, $F_R \leq 2.0894485 \cdot 10^{-8}$ at $N = 288$. Each leaf also records the F_R -inflated enclosure obtained by expanding the Taylor disk by F_R . The gate that this inflated disk excludes 0 certifies that the straight-line homotopy from D_N to $\det_H(I - L_{s,+})$ never vanishes on A . Consequently the winding of D_N along ∂Box_i equals the winding of the Fredholm determinant.

3.4. Winding. Argument increments are taken from a convex overlap-midpoint polygon homotopic to the image curve inside the nonzero inflated disks. The merged second-pin winding ball is

$$[0.999 \dots 996472261 \pm 1.22 \cdot 10^{-113}],$$

which pins the integer winding to 1. The first-pin winding ball likewise pins to 1. By the argument principle, $\det(1 - L_{s,+})$ has exactly one zero, counted with multiplicity, in each box. A zero of total multiplicity one is simple.

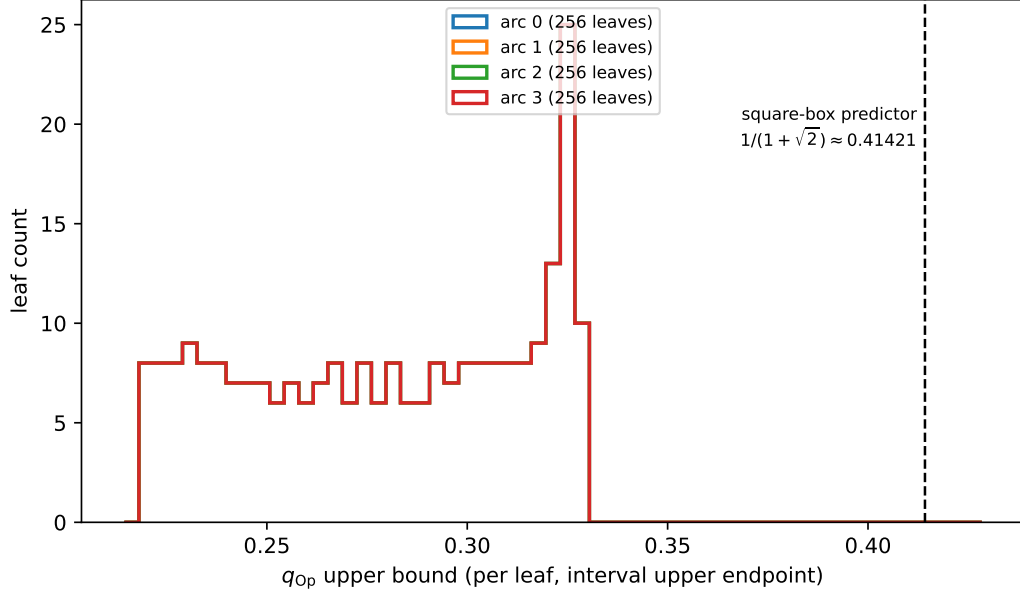


FIGURE 1. Per-arc distribution of q_{Op}^{upper} across 1024 depth-8 closed-contour checker leaves, arcs 0–3, with the square-box predictor $1/(1 + \sqrt{2}) \approx 0.41421$ marked. The recorded range is $[0.2188, 0.3273]$. The four series overplot.

3.5. What the receipts do not certify. The receipts do not evaluate Z_{G_5} or φ_5 . They do not certify that $\det(1 - L_{s,-})$ is nonvanishing in either box. They do not by themselves identify the Hilbert-space determinant with the MMS Banach determinant, nor do they resolve the printed $q > 5$ heading above [11, (34)]. Those steps are the open or cited joints of §5.

Figure 1 records the finite-section operator-norm bound q_{Op}^{upper} on 1024 depth-8 closed-contour checker leaves, four arcs of 256 leaves. Every value lies in $[0.2188, 0.3273]$, strictly below the square-box predictor $1/(1 + \sqrt{2}) \approx 0.41421$. The four per-arc histograms overplot.

4. TWO CERTIFIED ZEROS OF $\det(1 - L_{s,+})$

Theorem 4.1 (computer-assisted). *Let $L_{s,+}$ be the plus-sector transfer operator of G_5 in the sense of §2.2. The Fredholm determinant $\det(1 - L_{s,+})$ has zeros s_1 and s_2 of multiplicity one, with*

$$\Re s_1 \in [0.45389418007494470, 0.45389618007494470],$$

$$\Im s_1 \in [5.7635362417301305, 5.7635382417301305];$$

$$\Re s_2 \in [0.41054273549473627, 0.41054473549473627],$$

$$\Im s_2 \in [7.81976724701551188, 7.81976924701551188].$$

The closed real-part intervals are disjoint, with separation 0.04334944458020843.

Proof. The first-pin receipt `certificates/pin1_flagship/R3B_FLAGSHIP_CERT_RECEIPT.json` certifies winding 1 for D_{160} on ∂Box_1 , with every finite Taylor enclosure and every F_R -inflated enclosure excluding zero, minimum certified margin (finite lower bound minus F_R) at least $3.43786 \cdot 10^{-8}$, and $rH \leq 0.359 < 1$. The homotopy of §3 transfers the winding to $\det_H(I - L_{s,+})$. The argument principle therefore places exactly one zero of $\det(1 - L_{s,+})$, counted with multiplicity, in Box_1 . The second-pin merged receipt `chunk_receipts/S2_MERGED_CONTOUR_RECEIPT.json` does the same for $N = 288$ on ∂Box_2 : sixteen chunks cover the 192 base arcs, 452 accepted leaves,

merged winding 1, minimum certified margin at least $3.06455 \cdot 10^{-8}$, and $F_R \leq 2.0894485 \cdot 10^{-8}$. Multiplicity one is the winding. The displayed endpoints are the coordinate bounds of the two boxes. The closed-interval distance between the real-part projections is the exact decimal $0.45389418007494470 - 0.41054473549473627 = 0.04334944458020843$, re-verified with Arb directed rounding. $\square$

Winding 1 certifies a simple zero of $\det(1 - L_{s,+})$ only. If (1) applies at $q = 5$ and $\det(1 - K_s)$ is nonvanishing on the box, then Z_{G_5} vanishes at s_i with multiplicity at least one. Simplicity of the Z_{G_5} -zero would additionally require $\det(1 - L_{s,-})$ nonvanishing at s_i , which is not certified.

Remark. An $N = 128$ control on Box_2 fails the self-consistency gate ($rH > 1$ on a base arc). The $N = 288$ certificate is therefore not an artefact of a uniformly loose tail.

5. DEPENDENCY PROPOSITION

Theorem 4.1 is unconditional on the certificates of §3. The scattering interpretation — two zeros of φ_5 with distinct real parts in $(1/2, 1)$, hence no single vertical line $\Re s = c$ containing every nonreal strip zero of φ_5 — uses further joints. Each item below is marked CERTIFIED (an Arb receipt in the archive), CITED (a published theorem, with hypotheses checked in-line), or OPEN.

Proposition 5.1. *The passage from Theorem 4.1 to two zeros of φ_5 with distinct real parts in $(1/2, 1)$ factors as follows.*

- (J1) *Exact $q = 5$ plus-sector operator and determinant normalization: the operator of §2.2, sign +1, head split 4, Hardy direct sum, as implemented by the archived evaluation engine consumed by both receipts. CERTIFIED.*
- (J2) *Whole-contour Arb enclosure of D_N and of the F_R -inflated disks, with zero excluded on every accepted subarc of ∂Box_i . CERTIFIED.*
- (J3) *Winding equal to 1 on each contour. This certifies a simple zero of $\det(1 - L_{s,+})$ only. CERTIFIED.*
- (J4) *Finite section to the infinite Hilbert-space determinant: the identity $D_N = \det_H(I - L_{s,+}P_N)$ together with the Gohberg–Krein/Simon bound and the enclosed constants $T_{\text{tail}}(N)$, $B_{\text{same}}(N)$, F_R . CITED+CERTIFIED.*
- (J5) *Straight-line homotopy and the argument principle on ∂Box_i , using (J2) and (J4). CITED+CERTIFIED.*
- (J6) *Equality of the Hilbert-space determinant and the MMS Banach determinant on Ω^* in (2) (“Hilbert-to-Banach transport”). Both boxes lie in $\{\Re s > 0, \Im s > 1\}$. The identification is recorded as a paper-proof in the archive and is not reproduced here. OPEN.*
- (J7) *Exclusion of every zero of $\det(1 - K_s)$ from each closed box. By (4) those zeros have real part ≤ 0 , while both boxes satisfy $\Re s > 0$. CITED.*
- (J8) *The factorization (1) at $q = 5$, i.e. that [11, Theorem 6.4] applies despite the printed heading “ $q = 2h_q + 3 > 5$ ” above [11, (34)]. The general odd- q incidence formulas and the explicit $q = 5$ functional equation of that paper support the identification; the heading conflict is not dissolved in print. OPEN.*
- (J9) *A zero of $\det(1 - L_{s,+})$ is a zero of Z_{G_5} of multiplicity at least that of the plus factor. This follows from (1) once (J7) and (J8) hold, using holomorphy of $\det(1 - L_{s,-})$ near each box [11, Theorem 4.10]. Simplicity of the Z_{G_5} -zero is not claimed. CITED, conditional on (J8).*
- (J10) *A nonreal left-strip zero of Z_{G_5} is a pole of φ_5 of the same order, with no completed, trivial, or gamma divisor in either box: Theorem 2.1, item 6 of [3, §2.5], one-cusp scalar specialization. CITED.*
- (J11) *The functional equation $\varphi_5(s)\varphi_5(1-s) = 1$ reflects a pole of order m at s_i to a zero of order m at $1 - s_i$. No conjugation enters. CITED.*

(J12) *Transport of the two rectangles under $s \mapsto 1 - s$ and distinctness of the real parts. The image intervals are*

$$\begin{aligned}\Re \rho_1 &\in [0.54610381992505530, 0.54610581992505530], \\ \Im \rho_1 &\in [-5.7635382417301305, -5.7635362417301305];\end{aligned}$$

$$\begin{aligned}\Re \rho_2 &\in [0.58945526450526373, 0.58945726450526373], \\ \Im \rho_2 &\in [-7.81976924701551188, -7.81976724701551188],\end{aligned}$$

with the same closed-interval separation 0.04334944458020843. Both real-part intervals lie in $(1/2, 1)$. CERTIFIED.

(J13) *Membership of $(\varphi_5, \mathcal{D}_5)$ in a generic class of scattering pairs. That membership is not used in this paper. OPEN.*

Joints (J1)–(J5) and the s -box arithmetic in (J12) are the support of Theorem 4.1. The scattering and no-single-line consequence below is conditional on every CITED or OPEN joint among (J6)–(J12), and is thereby severed from the unconditional two-pin theorem. Joint (J13) is not consumed.

Corollary 5.2 (conditional on (J6)–(J12)). *Assume joints (J6)–(J11). Then Z_{G_5} has zeros at s_1 and s_2 of multiplicity at least one, φ_5 has poles at s_i and zeros $\rho_i = 1 - s_i$ of the same order, and the real-part intervals of ρ_1 and ρ_2 are those displayed in (J12). In particular, the nonreal zeros of φ_5 in $1/2 < \Re s < 1$ lie on no single vertical line $\Re s = c$.*

The last clause is elementary once two zeros with distinct real parts are granted: for any $c \in \mathbb{R}$, at least one of the two closed real-part intervals excludes c . Joint (J8) remains open, so Corollary 5.2 is not a theorem of this paper.

6. RELATED WORK

The novelty claim is narrow: these are, to our knowledge, the first *interval-certified* localizations of zeros of the G_5 plus-sector transfer-operator determinant, as against the heuristic numerics of [18]. The analytic apparatus is classical. Table 1 records, for each load-bearing predecessor, what it proves and what the present paper adds.

Fried [4] is the historical source of dynamical zeta functions as Fredholm determinants; it is not a finite-area Hecke cusp theorem. Borthwick–Judge–Perry [2] identify resonances with Selberg-zeta divisors for geometrically finite surfaces, typically in infinite area.

An open question. Does the fixed- q statement “for every integer $q \geq 3$, the scattering determinant φ_q has infinitely many nonreal zeros in $\Re s > 1/2$ ” follow as an immediate corollary of Selberg [16], Hejhal [6], and Kelmer [10] in orbifold generality? Hejhal’s Theorem 7.11 gives zeros in prescribed rectangles for all sufficiently large q , not for each fixed q . Kelmer’s global setup is torsion-free. We do not claim the fixed- q growth law as a theorem, and we do not claim to have settled this comparison.

7. MACHINE VERIFICATION

The Lean 4 artifacts in the archive formalize elementary conditional implications between stated hypotheses. No analytic or spectral content is formalized: there is no Lean definition of φ_5 , of $L_{s,+}$, of Z_{G_5} , or of a Hecke triangle group, and no theorem of MMS or FJS is in Lean.

The file `lean/v34_TwoPinNoLine/TwoPinNoLine.lean` proves four statements: an abstract set of complex numbers containing two nonreal right-strip members with distinct real parts admits no vertical line through all such members; the exact rational gap $4334944458020843/10^{17}$ between the two pin intervals; disjointness of those intervals; and distinctness of real parts for any pair of reals lying in the two intervals. The zeros are members of an abstract set. Multiplicity is not present:

TABLE 1. Theorem delta. “Adds” means relative to the cited result, not a claim of priority against the rest of the table.

Source	Proves	This paper
Selberg [16]	Weighted distribution of poles of Eisenstein series on hyperbolic surfaces; Jensen/Littlewood counting engine.	Not used. No counting theorem is claimed.
Hejhal [6]	Thm. 7.11 / Cor. 7.12: for all sufficiently large N , zeros (resp. poles) of φ_N in any prescribed rectangle touching $\Re s = 1/2$. Memoirs 469 [7]: numerical Hecke spectra.	Certificates at the single group G_5 , with explicit boxes; no large- q statement.
Kelmer [10]	Higher-dimensional distribution of Eisenstein poles; surface case attributed to Selberg. Thm. 3 and §4.	Not used.
Phillips–Sarnak [13, 15, 14]	Deformation of the augmented spectrum; cusp-form dissolution into resonances under a Fermi golden rule; spectrum of Hecke groups.	No deformation is used. G_5 is a fixed orbifold.
Garbin–Jorgenson [5]	Quantitative elliptic degeneration of the Hecke family; reproduces Hejhal 7.11/7.12; spectral counting $c_0(T) \log Q + O((\log Q)^{3/4})$.	No $q \rightarrow \infty$ statement.
Strömberg [18]	Numerical zeros of Z_{G_q} for $q = 3, 4, 5$ from truncated transfer operators; explicitly non-rigorous.	Interval certificates for two G_5 zeros of $\det(1 - L_{s,+})$, winding 1.
MMS [11]	Nuclear transfer operator of G_q ; factorization (1); meromorphy of the sector determinants.	Evaluates $\det(1 - L_{s,+})$ in ball arithmetic at $q = 5$. The $q > 5$ heading is left open (J8).
Möller–Pohl [12]	Period functions for every cofinite G_q ; Selberg zeta as a Fredholm determinant of accelerated transfer operators.	Different operator; not evaluated here.
Adam–Pohl [1]	Laplace eigenfunctions versus zeros of Z_{G_q} via slow/fast transfer operators, finite and infinite area.	Not evaluated here.
FJS [3]	Divisor of φ and of Z_Γ for cofinite orbifolds with unitary representations (Theorem 2.1).	Cited for the conditional promotion (J10), not re-proved.
Jorgenson–Smajlović [9]	Zeros of the derivative of the Selberg zeta on finite-volume surfaces.	Context only; not consumed.

the formalization does not distinguish a simple zero from a multiple zero, and it does not treat the Selberg-zeta multiplicity lower bound of (J9).

The file `lean/v33_Scat1Lemma31Reflection/Scat1Lemma31Reflection.lean` proves that if a meromorphic function on an open set stable under $s \mapsto 1 - s$ satisfies $\varphi(s)\varphi(1 - s) = 1$ away from poles, then a pole of meromorphic order $m \geq 1$ at s_* implies a zero of meromorphic order m at $1 - s_*$. Multiplicity is the mathlib meromorphic order (a pole of order m is order $-m$). The hypotheses are abstract; φ_5 is not instantiated.

Both files are sorry-free and depend only on the standard classical axioms `propext`, `Classical.choice`, and `Quot.sound`. They are regression checks for the elementary steps in (J11) and (J12), not independent evidence for Theorem 4.1.

8. DATA AVAILABILITY

Supporting material will be deposited at [DOI pending]. Until that deposit, the archive consists of the files named below, with paths relative to the archive root. No statement of archival completeness is made in advance of the DOI.

- `certificates/pin1_flagship/R3B_FLAGSHIP_CERT_RECEIPT.json` and `R3B_FLAGSHIP_CERT.md`: first-pin closed-contour certificate, $N = 160$, winding 1.
- `chunk_receipts/S2_MERGED_CONTOUR_RECEIPT.json`: merged second-pin closed-contour certificate, $N = 288$, `merged_winding=1`, together with the sixteen chunk receipts `S2_CHUNK_a000-012.json`, ..., `S2_CHUNK_a180-192.json`.
- `S2_MERGED_REFEREE_FABLE.md` and `S2_MERGED_REFEREE_SOL.md`: the two merged-contour referee reports.
- `lean/v34_TwoPinNoLine/TwoPinNoLine.lean` and `lean/v33_Scat1Lemma31Reflection/Scat1Lemma31Reflection.lean`, with `lean/VERIFICATION.md`.
- `fig1_qop_hist.pdf`, reproduced from `shard_receipts_d8/` by `make_fig1.py`.

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